MATHEMATICAL ASSESSMENT

Phase I — Foundations of Earthquake Processes, Observations, and Hazard Interpretation

Visiting Expert Program HRDP (DOST – PCIEERD)

This is a sample document to draft for Training Outputs and Assessment.

IMPORTANT: WHEN DRAFTING THE ASSESSMENT AND OUTPUT KEEP IN MIND (1) ANALYZE AND ANTICIPATE WHAT YOU WILL GET FROM THE TRAINING SESSION THAT COULD BE APPLIED IN YOUR RESPECTIVE DISCPLINE (MATH/CS/DS/EARTH SCI) - THIS WILL LATER SUPPORT THE SCIENTIFIC REVIEW ON THE PLATFORM DEVELOPMENT, (2) TRAINING OUTPUTS ARE DIFFERENT ON CS AND DS. CS WILL FOCUS ON FEATURES AND IMPLEMENTATION, DS WILL FOCUS ON DATA.

LINKS:

(1) **SESSION DESIGN AND OUTLINE.pdf This will be the PDF reference on the content that will be given in the session. The Interdisciplinary Relevance how you will draft the training output based on CS/DS**

Interdisciplinary Relevance

Discipline	Interdisciplinary Relevance
Mathematics	Participants in mathematics will explore the role of spatial reasoning, geometric interpretation, and attenuation behavior in understanding how earthquake effects vary across locations. Discussions will introduce the concept of uncertainty arising from incomplete observations and varying intensity distributions.
Data Science	Participants in data science will examine earthquakes as noisy and incomplete datasets. The session introduces challenges such as inconsistent reporting, observational bias, and variability in historical and instrumental records, emphasizing the importance of data interpretation rather than purely numerical processing.
Computer Science	Participants in computer science will be introduced to how earthquake information may be represented digitally through mapping systems, seismic databases, and signal representation frameworks. The session establishes the conceptual foundation for later discussions involving visualization and computational implementation.
Earth Science	Participants in earth science will focus on the tectonic and physical mechanisms that generate earthquakes and influence seismic

(2) **SCHEDULED PLAN OF ACTIVITIES UPDATED MAY.pdf This document shows that specific dates for the training. EACH SESSION HAS SPECIFIC OUTPUTS, THIS SHOULD BE REFLECTED IN YOUR DRAFT.**

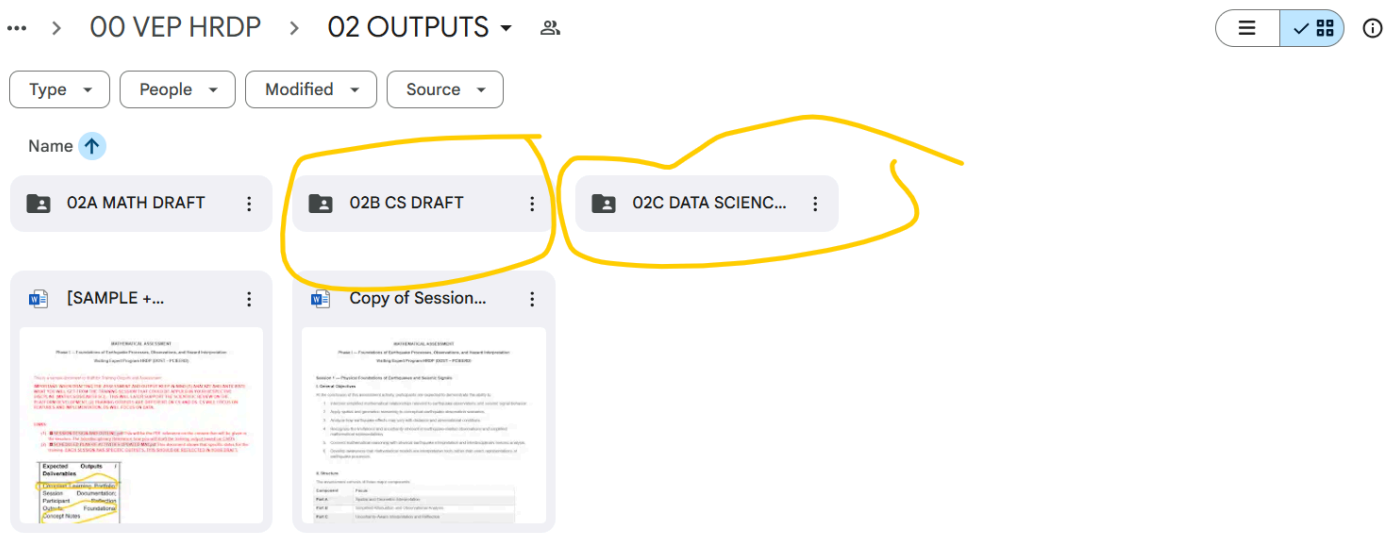
Expected Outputs / Deliverables
Compiled Learning Portfolio; Session Documentation; Participant Reflection Outputs; Foundational Concept Notes

Date	Session	Mode	Duration	Focus	Key Activities	To accomplish	Tangible Output
May 23, 2026	1	Online	2 hours	Physical Foundations of Earthquakes and Seismic Signals	Expert discussion on seismic waves, earthquake source mechanisms, wave propagation, intensity observations, physical damage, and observational uncertainty. Guided interdisciplinary discussion on earthquakes as measurable and inferential scientific phenomena.	Establish a foundational understanding of earthquake processes, observational systems, and uncertainty in seismic interpretation.	Compiled Learning Portfolio; Reflection Notes; Session Documentation

OTHER STUFFS:

1. **Each succeeding session is expected to be more difficult than the succeeding one, NOTHING SHOULD REPEAT.**
2. **DATA SCIENCE OUTPUTS could use using synthetic data sets or actual data sets.**
3. **Use this reference SAMPLE IN DRAFTING IN YOUR OWN TRAINING OUTPUTS. This one document is for the mathematics.**
4. **ckpobeso@addu.edu.phand kdptan@addu.edu.phwill draft the CS and zlfsulit@addu.edu.ph izomamac@addu.edu.phwill draft the Data science**
5. **Upload in your respective folders:**

   Folder Link: 02 OUTPUTS



Session 1 — Physical Foundations of Earthquakes and Seismic Signals

I. General Objectives

At the conclusion of this assessment activity, participants are expected to demonstrate the ability to:

1. Interpret simplified mathematical relationships relevant to earthquake observations and seismic signal behavior.
2. Apply spatial and geometric reasoning to conceptual earthquake observation scenarios.
3. Analyze how earthquake effects may vary with distance and observational conditions.
4. Recognize the limitations and uncertainty inherent in earthquake-related observations and simplified mathematical representations.
5. Connect mathematical reasoning with physical earthquake interpretation and interdisciplinary seismic analysis.
6. Develop awareness that mathematical models are interpretative tools rather than exact representations of earthquake processes.

II. Structure

The assessment consists of three major components:

Component	Focus
Part A	Spatial and Geometric Interpretation
Part B	Simplified Attenuation and Observational Analysis
Part C	Uncertainty-Aware Interpretation and Reflection

III. Assessment Proper

PART A: Spatial and Geometric Interpretation

Activity 1: Spatial Distribution of Earthquake Observations

A shallow earthquake was reported in Eastern Mindanao. Four municipalities reported varying felt intensities.

The approximate coordinates (in kilometers relative to a local reference system) are shown below:

Municipality	Coordinate (Lat, Long)	Reported Intensity
Municipality A	(7.3214, 126.0212)	VII
Municipality B	(7.0213, 125.3213)	VI
Municipality C	(7.2124, 126.2121)	V
Municipality D	(7.6521, 126.5014)	VI

Guide Questions

1. Using the Euclidean distance formula,
    - Municipality A and Municipality B
    - Municipality A and Municipality D
    - Municipality B and Municipality C

2. Based on the intensity distribution and spatial arrangement:
    - Which municipality may be closer to the possible source region?
    - Which observations appear spatially consistent?
    - Which observations may reflect local amplification or observational variability?

    Explain briefly.

3. Discuss why spatial proximity alone may not perfectly determine earthquake intensity.
    Your discussion may consider:
    - local geological conditions,
    - building types,
    - observational subjectivity,
    - and incomplete observations.

PART B: Simplified Attenuation and Observational Analysis

Activity 2: Conceptual Attenuation of Earthquake Effects

A simplified conceptual attenuation relationship is given below:

$$I(d) = I_0 - kd$$

where:

- $(I(d))$ = estimated intensity at distance (d),

- (I_0) = source intensity,
- (k) = attenuation constant,
- (d) = distance from source.

Assume:

- ($I(d)$ = 8)
- (k = 0.08)

Tasks

1. Estimate the intensity values at:
    - 10 km
    - 20 km
    - 35 km
    - 50 km

    from the conceptual source.
2. Construct a simple table showing:

Distance	Estimated Intensity

3. Describe the observed trend. What happens to estimated intensity as distance increases?
4. Explain why real earthquake attenuation may not follow a perfectly linear relationship.

    Your answer may consider:

- geological variability,
- wave scattering,
- soil conditions,
- and observational limitations.

PART C: Uncertainty-Aware Interpretation and Reflection

Activity 3: Interpreting Incomplete Observations

A historical earthquake (American Colonial Period) account from a coastal Philippine settlement state:

"*The shaking was strong enough to crack church walls and cause objects to fall from shelves. Nearby coastal communities reported weaker effects, but no precise measurements existed.*"

Tasks

1. Identify at least three forms of uncertainty that may arise when interpreting this account.

Examples may include:

- observational subjectivity,
- incomplete reporting,
- uncertain distance,
- unknown construction quality,
- and missing intensity data.

2. Suppose two nearby towns reported different levels of damage despite being approximately the same distance from the earthquake source.

   Provide at least two possible explanations.

3. Explain why historical earthquake interpretation is considered an inverse and inferential problem rather than a direct measurement problem.

   Your explanation should connect:

   - observations,
   - physical processes,
   - uncertainty,
   - and interpretation.

IV. Synthesis Activity

Integrated Reflection

Write a short synthesis reflection (approximately 300–500 words) addressing the following:

“How do mathematical reasoning and observational uncertainty interact in earthquake interpretation?”

Participants are encouraged to discuss:

- spatial interpretation,
- attenuation concepts,
- uncertainty,
- observational limitations,
- and the role of mathematics in supporting interdisciplinary earthquake analysis.

The reflection should emphasize that mathematical representations are interpretative tools that operate within incomplete and uncertain observational environments.

V. Expected Outputs

Participants are expected to submit the following:

Output	Description
Spatial Interpretation Worksheet	Solutions and interpretation for Part A
Attenuation Exercise Table	Computations and observations for Part B
Uncertainty Reflection Responses	Written responses for Part C
Integrated Reflection Paper	Final synthesis reflection